Grant Proposal: Private Lending on Aleo

PWN DAO Foundation · Aleo SCALE grant programme · 2 November 2026
Request: $85,000 · Delivery: 17 weeks from project start
Open-source Aleo programs, TypeScript SDK, reference app, independent audit and mainnet launch.

Project Overview

PWN is building peer-to-peer lending on Aleo. Lenders and borrowers can post public or private offers, both allowing multiple fills. Each fill creates a separate loan with a fixed rate, duration and collateral requirement. Borrowers can repay until the lender's default claim settles. Loans use no price oracles or price-triggered liquidations.
The grant takes our proof of concept to an audited mainnet release, starting with ALEO collateral and USDCx credit. Programs, SDK and reference app will be released under GPL-3.0, with zero protocol fees in V1.

Team

PWN DAO Foundation has built crypto lending infrastructure for over five years. PWN is deployed on 13 EVM mainnets and has processed more than $10m in loan volume for over 5,000 users. The team also builds OWN, crypto-backed mortgage infrastructure using PWN's lending primitives.

- **Josef Jelacic** — founder of PWN DAO and OWN; previously at the Ethereum Foundation. Protocol architecture and design oversight. LinkedIn.
- **Šimon Kozák** — founding engineer and CTO at PWN DAO and OWN. Protocol and application development. LinkedIn · GitHub.
- **Jan Mikšík** — engineer with four years of experience building DeFi lending applications. Protocol and application development. LinkedIn · GitHub.

Šimon and Jan will jointly deliver the Leo programs, SDK and reference app.

Market and Ecosystem Fit

Aleo has USDCx, USAD and liquid-staking assets. The protocol adds collateralized credit to those assets, with three initial uses:

- **Liquidity for asset holders:** borrow against ALEO without selling. Follow-on integrations would let holders borrow against pALEO and stALEO while retaining staking exposure.
- **Lending for stablecoin holders:** set a rate, term and collateral requirement, or accept a borrower's offer. Lenders bear default and collateral-price risk.
- **Private bilateral credit:** funds, DAOs and treasuries share offers with chosen counterparties while limiting public links between their wallets and loans.

On 2 November 2026, USDCx and USAD had combined supply of about $4.71m at peg. Galaxy reported $56.16bn in crypto-collateralized loans at Q2 2026, including $20.43bn in DeFi; Federal Reserve researchers estimated $138bn in US securities-based loans in Q1 2024. These figures describe available assets and adjacent markets, not demonstrated demand or a combined TAM. Pilot participants and completed loans will validate demand.

The SDK lets wallets and payment applications embed these flows. Integration targets include Shield wallet, USDCx, USAD, Pondo and Beta Staking; these are planned integrations, not confirmed partnerships. Public offer discovery and settlement verification work from Aleo state, with optional indexing for faster search.

The work fits the Aleo SCALE grant programme's DeFi and open-source scope: reusable lending programs, an SDK, integration documentation and published security findings.

System Design

Offers and loan lifecycle

Both offer directions set assets, total capacity, minimum and maximum fill amounts, collateral and repayment pricing, duration and expiry. Public offers can serve different counterparties; private offers designate one counterparty who may fill repeatedly. Each fill binds its own lender, collateral and maturity.

Lender posts an offer Credit escrowed · Borrowers fill	**Borrower posts an offer** Collateral escrowed · Lenders fill
Public listing or private invitation · Repeated fills ↓	
Each fill creates a separate loan Collateral allocated · Principal delivered ↓	
Repayment settles first Lender receives repayment Borrower recovers that loan's collateral	**Claim settles first, at/after maturity** Loan closes Lender receives that loan's collateral

Repayment remains possible until a lender's default claim settles. Claims are allowed at or after maturity; maturity does not transfer collateral automatically. The first successful settlement closes only that loan.

Funding baseline. The maker escrows credit for a lender offer or collateral for a borrower offer. The accepting party supplies the other asset at loan creation. Transfers and accounting succeed together or revert. Expiry blocks new fills; cancellation refunds unused escrow without changing active loans.

Both directions support repeated fills within remaining capacity. Borrower-offer repayments belong to the matched lender of each loan. Repayment and collateral recovery do not replenish offer capacity. Collateral rounds upward per loan; borrower fills must fit remaining requested principal and reserved collateral. Unused remainders are refundable.

Funding at acceptance: M1 feasibility check. A local USDCx allowance probe funded credit and native collateral atomically at acceptance; quote posting and approval moved no principal or collateral. This exposes the lender's public funding address and amount, leaves balances unreserved, and requires handling revocation or insufficient funds. Native private collateral still needs borrower authorization at final execution; an offer signature alone cannot spend another party's records. M1 validates live token and wallet support and publishes the result. Funded offers remain the baseline. Local result · Aleo transfer model.

Modules and custody

- **Hub and Config:** authorize offer modules and stop new originations. Production controls cannot redirect or block settlement of existing loans.
- **Offer modules:** validate terms and acceptance rights; manage lender credit, cancellation and withdrawals; and keep borrower-offer repayments separate for each matched lender.
- **Loan core:** holds unused borrower collateral reservations and active collateral, creates loans, and enforces repayment, default claims and one-time settlement.
- **Token programs:** execute transfers. Funds in custody use program-controlled public balances; user payouts use private records. Programs never hold spendable private records.
- **SDK and app:** prepare transactions, show remaining capacity and individual loans, manage records and backups, discover public offers, and share/import encrypted payloads. They verify openings against chain commitments.

V1 has no protocol asset allowlist or administrative admission. Assets must match supported interfaces; settlement binds the exact token program and accounting is isolated per token. Module authorization does not approve assets. Interface compatibility does not establish token trustworthiness.

Production starts with ALEO collateral and USDCx credit. The PoC covers native ALEO, ARC-20 and USDCx/USAD profiles; pALEO and stALEO are follow-on integrations. Issuer freeze, burn, pause and upgrade powers may disrupt settlement. M1 defines token-deficit handling. NFTs, rebasing and fee-on-transfer tokens are outside V1.

Privacy

V1 aims to keep participant wallet addresses out of public lending fields. Private offers bind a designated counterparty and supplied terms to a salted on-chain commitment; that counterparty may fill repeatedly. Complete payloads are shared privately. Encrypted delivery, import and recovery are SDK deliverables for both offer directions.

Data	Visibility and limits
Participant addresses and ownership	Private inputs, user-owned records and salted authorization avoid direct public wallet-to-position fields. Counterparties receive necessary payout details; correlation and counterparty disclosure remain possible.
Public offers	Advertised terms, remaining capacity and status are public. Public borrower acceptance also needs an off-chain opening with payout details; once shared, the maker can be offline.
Private offers	Complete payloads and the designated counterparty are shared privately. Commitments, expiry, status, necessary capacity data and escrow movements remain visible; escrow can reveal assets and amounts before acceptance.
Active loans and custody	Submitted loan amounts, collateral, rate and duration are public even if acceptance is rejected. Created loans also expose maturity, status and program balances. Confidential settlement and hidden amounts are outside V1.
Compliance and proving	USDCx/USAD compliance observers can read designated transfer receipts. A delegated prover may see private inputs unless protected by a verified confidential proving setup.
Selective disclosure	The SDK will support disclosing one loan without sharing an account-wide view key. Counterparties still receive the payout details needed for that loan.

The SDK requires private fees without silent public-fee fallback, fresh ownership and opening salts, encrypted backups and recovery. The privacy review covers successful and rejected transactions, indexers, delegated provers and timing/amount correlation.
References: PoC disclosure matrix · Aleo records and view keys · delegated proving · stablecoin compliance visibility.

Current Implementation

The public-testnet predecessor executed funding, draw, repayment, default claim, refund and withdrawal with real proofs, using native ALEO credit and token-registry collateral. The multi-token PoC revision implements multiple fills in both directions, private commitments and no asset admission. Local lifecycle, accounting, adversarial and arithmetic tests pass. Selected transactions generate proofs, but the devnode does not verify them or run consensus.

Proof-enabled public-network validation, live issuer and wallet integration, encrypted delivery and recovery remain grant work.
The production core will be audited and deployed without upgradeability, replacing PoC timelocked upgrades. Installments, refinancing and pooled crowdfunded loans are outside V1.
Evidence: PoC interfaces · PoC security review · Offer semantics · Validation results · compliant stablecoin programs.

Funding and Milestones

The $85,000 request covers 17 weeks of development, integration, testing and independent audit. Each milestone is verified through published code, test results and deployment evidence.

Milestone	Completion	Amount
M1 · Core and interfaces	Week 7	$35,000
M2 · SDK and offer discovery	Week 10	$15,000
M3 · Reference app	Week 13	$15,000
M4 · Audit and launch	Week 17	$20,000
Total	**17 weeks**	**$85,000**

M1 · Core and interfaces

Deliver: production programs with multiple fills in both offer directions and visibility modes; Hub/Config; supported token interfaces without asset admission; live USDCx integration; custody and settlement interfaces; threat and disclosure models. Publish the allowance feasibility result and freeze reviewed interfaces.
Acceptance: on public testnet, one lender offer serves two borrowers and one borrower offer is funded by two lenders. Designated private counterparties fill repeatedly; unauthorized takers are rejected. Run a live USDCx lifecycle and document compliance outputs. Proof-enabled, property and fuzz tests cover allocation and rounding, reservation limits, cancellation, expiry, overfill, modified terms, replay, premature claims, settlement races, administration and token failures. Publish privacy findings for successful and rejected transactions.

M2 · SDK and offer discovery

Deliver: TypeScript SDK for offer creation and fills, public discovery, encrypted payload sharing/import, repayment, claims, cancellation and withdrawals. Include private fees, current USDCx compliance proofs, capacity checks and position recovery.
Acceptance: publish the package and integration docs; complete the testnet lifecycle without a proprietary backend; verify imported openings against commitments; recover positions from backup; and report stale compliance proofs or failed transactions with actionable errors. Private payloads are not publicly indexed.

M3 · Reference app

Deliver: wallet connection, both offer directions, public browsing, private invitations, fill sizing, remaining capacity, individual loan positions, repayment, claims and selective disclosure.
Acceptance: external testers complete lender and borrower journeys without team assistance, including private offers. Demonstrate disclosure of one loan without an account-wide view key; publish usage docs and fix blocking usability findings. Agree tester count, supported wallets and usability criteria with Aleo before M3 begins.

M4 · Audit and launch

Deliver: the $20,000 milestone covers independent audit of the core and integration boundaries, remediation, retesting, immutable mainnet deployment and final SDK release. Audit preparation starts after M1.
Acceptance: publish the audit and remediation status, resolving material findings before mainnet deposits. Publish program IDs, source/build references, constructor policies and operational docs. Complete at least one mainnet loan with external participants through origination, repayment and settlement.
If a material external dependency blocks delivery, agree a revised scope and schedule with Aleo.

Six-month targets after launch

50 loans originated; $250,000 in cumulative loan principal, measured in USD across supported assets; and two third-party apps integrating the SDK.